Cambridge International Education

Cambridge IGCSE™

DESIGN AND TECHNOLOGY **0445/31**

Paper 3 Resistant Materials **May/June 2026**

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

This document consists of **10** printed pages.

General marking principles

All examiners must apply these marking principles when marking candidate responses.

1	Examiners must award marks for each response in line with: • the specific **content** defined in the mark scheme • the specific **skills** defined in the mark scheme or in the level descriptions • the **standard of response** required as exemplified by the standardisation scripts.
2	Examiners must award **whole marks** (not half marks, or other fractions).
3	Examiners must award marks **positively**. This means that: • marks are not deducted for errors or omissions • marks are awarded for correct/valid answers as defined in the mark scheme and also for **other valid answers** which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate • the quality of spelling, punctuation and grammar are judged **only** when the mark scheme indicates that these features are **specifically assessed** in the question. However, the meaning of all responses must be unambiguous.
4	Examiners must **consistently** apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.
5	Examiners must use the **full range of marks** defined in the mark scheme, unless limited by the quality of the candidate responses seen.
6	Examiners must award marks according to the mark scheme and **must not** award marks with grade thresholds or grade descriptions in mind.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
?	Unclear
BOD	Benefit of the doubt
✗	Incorrect point
ECF	Error carried forward
N/A	Highlighting areas of text
NBOD	No benefit of doubt given
N/A	Off-page comment – allows comments to be entered off the page
REP	Repeat
SEEN	Indicates that the point has been noted, but no credit has been given
SEEN	Indicates that the point has been noted, but no credit has been given (big)

Annotation	Meaning
✓	Correct point
TV	Too vague
DET	Relevant detail

Question	Answer	Marks	Guidance
1	Any 2 specification points: number of toothbrushes to hold, sizes, moisture resistant, materials, attractive appearance, ease of access, easy to clean, stable in use, durable, no sharp corners, take up little space, corrosion resistant [2 × 1]	**2**	**Do not accept** lightweight, aesthetically pleasing Accept any valid points

Question	Answer	Marks	Guidance
2	Length of tenon at least ½ width of upright [1] Rail **A** divided into 3 equal parts across grain [1] Height of tenon to correspond with height of mortise [1]	**3**	Tenon could be long enough for a through mortise

Question	Answer	Marks	Guidance
3	**A** tool post [1] **B** tailstock [1] **C** bed [1]	**3**	

Question	Answer	Marks	Guidance
4	Clothes pegs injection moulding [1] Tubes extrusion [1] Model car vacuum forming [1]	**3**	Accept injection forming Accept vacuum moulding

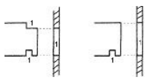
Question	Answer	Marks	Guidance
5	Steel insert shown in end of tube **or** applied to the end [1] Screw threaded hole shown in insert or noted [1]	**2**	Accept nut shown fixed in end

Question	Answer	Marks	Guidance
6	Softwood can expand and contract [1] Not gluing boards enables them to 'move' [1]	**2**	

Question	Answer	Marks	Guidance
7(a)	Pine	**1**	
7(b)	Plywood	**1**	

Question	Answer	Marks	Guidance
8(a)	Smoothing or jack plane	**1**	
8(b)	Bench stop or custom made 'stops', double sided tape [1] Softwood length shown in position [1]	**2**	Accept any practical method

Question	Answer	Marks	Guidance
9(a)	Tee hinge	**1**	
9(b)	Brace adds rigidity, prevents door from dropping / collapsing	**1**	**Do not accept** supports or strengthens door **Do reward** reference to triangulation

Question	Answer	Marks	Guidance
10	 'Slot' removed in drawer front [1] 'Slots'/rebates in handle [2 × 1]	**3**	

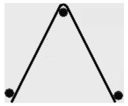
Question	Answer	Marks	Guidance
11(a)(i)	Purpose of veneer is to cover a manufactured board to enhance appearance	**1**	
11(a)(ii)	Drawbacks: easily damaged (e.g. chip, split, peel off) not as durable as solid wood, relatively thin veneer is difficult to repair	**1**	**Do not accept** scratches
11(b)(i)	K-D fittings: 1-piece and 2-piece corner blocks, rigid joint, cam lock **named** [1] Technical accuracy of sketch [0 – 2]	**3**	Screw holes must allow screws to fit into both pieces
11(b)(ii)	Minimum 2 dowels-maximum 4 dowels [1] Diameter of dowels minimum 6 mm – maximum 10 mm [1] Equally spaced [1]	**3**	Accept any number stated between 6 and 10 Appropriate to the number of dowels
11(c)(i)	The housing is not visible at the front of the magazine rack	**1**	
11cii	'Cut out' shown on **end** of shelf [1] 'Cut out' shown on **edge** of shelf [1]	**2**	
11(c)(iii)	2 safety precautions: wear eye protection, no loose clothing, trailing leads, check condition of tool, secure workpiece, long hair tied back, turn off power when not in use, read instructions [2 × 1]	**2**	Accept any valid precautions
11(d)(i)	2 benefits: quicker than by hand, more even / flat finish, more pressure can be applied, less effort [2 × 1]	**2**	**Do not accept** make smooth Accept any valid benefits
11(d)(ii)	Application of grades from coarse – fine [1] 'Scratches' created by one grade are replaced with finer 'scratches' [1]	**2**	Accept reference to different grades produce different end results

Question	Answer	Marks	Guidance
11(d)(iii)	Varnish, white polish, French polish, Danish oil, sealer, lacquer, shellac, linseed oil, wax [2 × 1]	**2**	**Do not accept** 'oil'
11(e)	20 mm 'foot' clearly shown [1] Constructional details [0 – 2]	**3**	**Do not accept** mortise and tenon joint
11(f)	Practical methods: Some sort of 'lipping' [1] Named materials [specific-**NOT** generic] [1] Constructional details [1] **OR** Back panel attached to the whole of the back [1] Named materials: hardboard, plywood, MDF back [1] Constructional details: e.g. use of pin and glue [1]	**3**	**Do not reward** solid wood as an appropriate material for a complete back panel

Question	Answer	Marks	Guidance
12(a)	2 advantages: cheaper, more stable, no need for separate parts to be joined [2 × 1]	**2**	Accept any general advantages: for example, more readily available, wide boards available
12(b)	Drill hole in plywood [1] Insert blade of named saw and cut out waste [1] File **or** glasspaper edges smooth and flat [1]	**3**	Reward sawing and filing/glasspapering even if a hole is not drilled Saws include: coping, Hegner, scroll, jig saw
12(c)	Methods of joining named: dowel, half-lapped, mortise and tenon, dovetail, finger, biscuit [1] Technical accuracy of joint [0 – 3]	**4**	Accept use of brackets
12(d)(i)	2 marking out tools: pencil, steel rule, marking knife, try square, marking gauge [2 × 1]	**2**	Accept engineer's try square
12(d)(ii)	Hole saw, tank cutter	**1**	

Question	Answer	Marks	Guidance
12(e)	Strip heater/line bender/hot air gun [1] Use of a jig or former [1] Method of retention while cooling: use of cramps, weights [1] Accuracy of named tools mark can be awarded if **ALL 3** stages are given [1]	**4**	**Do not accept** heated in an oven
12(f)	length of screw minimum 10 mm – maximum 30 mm [1] type of head round or countersunk/countersink [1] material brass, mild steel, stainless steel, titanium [1]	**3**	
12(g)(i)	Acrylic, ABS, polystyrene, polypropylene, polycarbonate, PVC, polyethylene, PP, HIPS	**1**	**Do not accept** LDPE
12(g)(ii)	If the plastic is not heated enough it will not form If the plastic is overheated it will stretch and possibly melt The plastic can take the shape of the mould [2 × 1]	**2**	
12(g)(iii)	Stages include: test plastic during heating, raise platen, switch on pump, leave to cool, lower platen [3 × 1]	**3**	Accept any valid ‘intermediate ‘stage’ **Do not accept** switch off heater

Question	Answer	Marks	Guidance
13(a)	Dimensions of guitars, is it portable, ease of access, does not damage the guitar when held [2 × 1]	**2**	Reward **one** reference to dimensions only Accept any valid items of research
13(b)	2 benefits: durable material, relatively cheap, easy to work, plentiful supply, wide variety of standard forms, relatively malleable, can be given variety of finishes [2 × 1]	**2**	Accept any valid benefits
13(c)(i)	To make it ‘softer’, easier to work	**1**	

Question	Answer	Marks	Guidance
13(c)(ii)	Metal is heated [1] Allowed to cool [1]	2	
13(d)	 award 1 mark for each correct 'rod' [3 × 1] Added notes: for example, diameter, material, length of rods used [1]	4	
13(e)	**A** blow torch, gas torch, brazing torch [1] **B** firebrick [1] **C** brazing rod, spelter, filler rod [1]	3	**Do not accept** bricks
13(f)(i)	Wet and dry [silicon carbide] paper, emery cloth, aluminium oxide [2 × 1]	2	**Do not accept** steel wool
13(f)(ii)	Dip coating, electroplating, powder coating, galvanising, chrome plating	1	
13(g)	Stage 1 glue veneers together [1] Stage 2 use of male and female formers [2 × 1] Stage 3 method of clamping [1]	4	Accept use of vacuum forming bag: Details include: veneers glued, use of former, bag sealed, pump to extract air turned on
13(h)	Area **A** use of a wooden block or 'soft' material attached -stated [1] sketch [1]	2	
13(i)	Area **B** some form of 'foot' to protect the floor -stated [1] sketch [1]	2	